



Database Schema & Data Dictionary

PostgreSQL Models, Enums, Constraints & Indexes

DATA ARCHITECTURE

Effective Date: September 16, 2026

System: Order Control System v1.0

Database Schema & Data Dictionary

Order Control System — Multi-Brand Dark Kitchen Operating Platform

Printable Edition: [Download Executive PDF \(A4 Edition\)](#)

Database Engine: PostgreSQL 15+ (Supabase) | **ORM:** Prisma 7 (@prisma/adapters-pg)

Schema Version: 2026.09 Production

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1. Overview & Data Architecture

The Order Control System persistence layer is modeled in PostgreSQL and managed via Prisma 7. The architecture enforces ACID transactional integrity, soft deletion (`isActive = false` or `status = 'CANCELLED'`), and immutable historical records.

Key Architectural Invariants

- No Direct Column for Total Revenue on Order:** The `Order` model explicitly avoids storing a mutable `total` column. Instead, `subtotal`, `discount`, and `deliveryFee` are stored as atomic primitives, and totals are computed dynamically via `calculateOrderTotals()` to guarantee mathematical consistency.
- Snapshot Pricing on OrderItem:** `OrderItem.unitPrice` captures the frozen unit price at the time of order creation. Future price edits on `Product` do not mutate historical order records.
- Advisory Locks for Sequencing:** Order sequence numbers (`ORD-YYYYMMDD-XXXX`) utilize PostgreSQL advisory locks (`pg_advisory_xact_lock`) to serialize concurrent creations.

2. Enumeration Types (Enums)

Role

Defines role-based permissions across API endpoints and UI views.

- OWNER** : Full system access, audit inspector, financial reports, user management.
- MANAGER** : Shift management, menu edits, expense tracking, discount approvals, daily closing.
- CASHIER** : Fast order entry, order status advancement, discount requests, shift open.

OrderStatus

Governs the deterministic, unidirectional state machine.

- NEW** : Order ingested into the system.
- CONFIRMED** : Customer verified by phone; ticket dispatched to kitchen.
- PREPARING** : Line chef actively preparing items; live timer ticking.
- READY** : Items cooked, packaged, boxed, and awaiting courier pickup.
- OUT_FOR_DELIVERY** : Dispatched with driver (driver assignment mandatory).
- DELIVERED** : Successfully handed over to customer (terminal state).
- CANCELLED** : Voided order with mandatory cancellation reason (terminal state).

PaymentMethod

- CASH** : Cash collected upon delivery.
- VISA** : Credit/debit card collected at doorstep via portable POS terminal.
- ONLINE** : Pre-paid order via aggregator platform or payment gateway.

DriverType

- OWN** : In-house delivery courier employed by the restaurant.
- APP** : Courier provided by third-party delivery platform (e.g. Talabat Go). *Invariant: Automatically zeros restaurant delivery fee.*
- EXTERNAL** : Independent third-party courier (e.g. InDrive, freelance).
- PICKUP** : Direct pickup (deprecated / internal testing).

DiscountStatus

- NONE** : No discount applied.
- PENDING** : Requested by cashier with mandatory reason; awaiting manager approval.
- APPROVED** : Confirmed by manager; deducted from order total.
- REJECTED** : Declined by manager; order reverts to full subtotal.

CancelReason

Mandatory categorization required when cancelling any order:

- `CUSTOMER_CHANGED_MIND` : Customer called back to cancel.
- `DELIVERY_ISSUE` : Address unreachable, severe weather, courier unavailable.
- `QUALITY_ISSUE` : Customer returned food or disputed preparation.
- `NO_ANSWER` : Customer phone unanswered or switched off upon arrival.
- `ITEM_UNAVAILABLE` : Raw ingredients or packaging out of stock.
- `OTHER` : Unlisted reason (requires explicit descriptive notes).

AuditAction

Categorizes audit log entries:

- `CREATE` : Creation of order, expense, customer, or user.
- `UPDATE` : Entity attribute modifications.
- `CANCEL` : Cancellation of order.
- `STATUS_CHANGE` : Progression through state machine.
- `DISCOUNT_REQUEST` : Cashier requested a discount.
- `DISCOUNT_APPROVE` : Manager approved discount.
- `DISCOUNT_REJECT` : Manager rejected discount.

3. Identity & Authentication Models (`User`)

Model: `User`

Represents staff members and administrative accounts.

Column	Type	Nullable	Default	Description
<code>id</code>	<code>UUID</code>	No	<code>gen_random_uuid()</code>	Primary Key (matches Supabase Auth user ID)
<code>email</code>	<code>String</code>	No	—	Unique login email address
<code>name</code>	<code>String</code>	No	—	Full employee name
<code>role</code>	<code>Role</code>	No	<code>CASHIER</code>	RBAC authorization role
<code>isActive</code>	<code>Boolean</code>	No	<code>true</code>	Soft deletion status (false prevents login)
<code>createdAt</code>	<code>DateTime</code>	No	<code>now()</code>	Account creation timestamp
<code>updatedAt</code>	<code>DateTime</code>	No	Auto	Last profile update timestamp

4. Master Data & Catalog Models

Model: `Brand`

Represents one of the 4 virtual sushi brands.

Column	Type	Nullable	Description
id	UUID	No	Primary Key
name	String	No	Brand name in English (e.g. "Flower Sushi")
nameAr	String	No	Brand name in Arabic (e.g. "سوشي فلور")
code	String	No	Unique identifier code (e.g. "FLOWER")
isActive	Boolean	No	Soft-activation status

Model: Platform

Represents an aggregator channel or direct ordering medium.

Column	Type	Nullable	Description
id	UUID	No	Primary Key
name	String	No	Platform name in English (e.g. "Talabat")
nameAr	String	No	Platform name in Arabic (e.g. "طلبات")
code	String	No	Unique identifier code (e.g. "TALABAT")
isActive	Boolean	No	Soft-activation status

Model: Category

Menu categories mapped to specific brands.

Column	Type	Nullable	Description
id	UUID	No	Primary Key
brandId	UUID	No	Foreign Key → Brand.id
name	String	No	Category name (e.g. "Fried Maki")
nameAr	String	No	Category Arabic name (e.g. "فرايد مكي")
sortOrder	Int	No	Display sorting position
isActive	Boolean	No	Soft-activation status

Model: Product

Individual sushi rolls, combos, beverages, and sides.

Column	Type	Nullable	Description
id	UUID	No	Primary Key
categoryId	UUID	No	Foreign Key → Category.id
name	String	No	Product name (e.g. "Philadelphia Roll 8 Pcs")
nameAr	String	No	Arabic name (e.g. "فيلادلفيا رول 8 قطع")
price	Decimal(10,2)	No	Standard selling price in EGP
isActive	Boolean	No	Soft-activation status

5. Fleet & Logistics Models

Model: DeliveryZone

Geographic delivery zones with flat delivery fees.

Column	Type	Nullable	Description
id	UUID	No	Primary Key
name	String	No	Zone name (e.g. "Heliopolis")
nameAr	String	No	Arabic zone name (e.g. "مصر الجديدة")
fee	Decimal(10,2)	No	Delivery fee charged to customer (EGP)
isActive	Boolean	No	Soft-activation status

Model: DeliveryDriver

Delivery couriers and fleet personnel.

Column	Type	Nullable	Description
id	UUID	No	Primary Key
name	String	No	Courier full name
phone	String	No	Courier contact phone number
type	DriverType	No	Fleet classification (OWN , APP , EXTERNAL)
isActive	Boolean	No	Active working status

6. Customer Relationship Management (Customer)

Model: Customer

Centralized customer directory keyed by normalized phone number.

Column	Type	Nullable	Default	Description
id	UUID	No	gen_random_uuid()	Primary Key
phone	String	No	—	Normalized Egyptian phone number (Unique)
name	String	No	—	Customer full name
address	String	Yes	—	Primary delivery address
notes	String	Yes	—	Customer notes & allergy advisories
totalOrders	Int	No	0	Aggregated completed orders count
lifetimeSpent	Decimal(10,2)	No	0	Aggregated total spend across all brands
lastOrderDate	DateTime	Yes	—	Timestamp of most recent order

7. Core Transactional Models (Order , OrderItem)

Model: Order

The primary operational entity governing food preparation and accounting.

Column	Type	Nullable	Default	Description
id	UUID	No	gen_random_uuid()	Primary Key
orderNumber	String	No	—	Unique sequential number (<code>ORD-YYYYMMDD-XXXX</code>)
platformId	UUID	No	—	Foreign Key → <code>Platform.id</code>
brandId	UUID	No	—	Foreign Key → <code>Brand.id</code>
externalId	String	Yes	—	External aggregator ticket ID (e.g. <code>TAL-88491</code>)
customerId	UUID	No	—	Foreign Key → <code>Customer.id</code>
zoneId	UUID	Yes	—	Foreign Key → <code>DeliveryZone.id</code>
driverId	UUID	Yes	—	Foreign Key → <code>DeliveryDriver.id</code>
cashierId	UUID	No	—	Foreign Key → <code>User.id</code> (Order creator)
shiftId	UUID	Yes	—	Foreign Key → <code>Shift.id</code>
status	OrderStatus	No	NEW	State machine progression status
paymentMethod	PaymentMethod	No	CASH	Payment method classification
subtotal	Decimal(10,2)	No	0	Sum of all order item subtotals (EGP)
discount	Decimal(10,2)	No	0	Deducted discount amount (EGP)
discountReason	String	Yes	—	Mandatory reason for discount request
discountStatus	DiscountStatus	No	NONE	Approval lifecycle status
discountApprovedById	UUID	Yes	—	Foreign Key → <code>User.id</code> (Approving manager)
deliveryFee	Decimal(10,2)	No	0	Applied delivery fee (0 if <code>DriverType</code> <code>≡</code> <code>APP</code>)
notes	String	Yes	—	Freeform kitchen preparation notes
cancelReason	CancelReason	Yes	—	Mandatory reason if order cancelled

Column	Type	Nullable	Default	Description
cancelNotes	String	Yes	—	Supplemental notes for cancellation
createdAt	DateTime	No	now()	Ingestion timestamp
confirmedAt	DateTime	Yes	—	Phone confirmation timestamp
preparingAt	DateTime	Yes	—	Kitchen prep start timestamp
readyAt	DateTime	Yes	—	Food boxed/packaged timestamp
dispatchedAt	DateTime	Yes	—	Handover to courier timestamp
deliveredAt	DateTime	Yes	—	Final delivery completion timestamp
cancelledAt	DateTime	Yes	—	Order voided timestamp

Model: OrderItem

Normalized line items linked to an Order.

Column	Type	Nullable	Description
id	UUID	No	Primary Key
orderId	UUID	No	Foreign Key → Order.id (CASCADE on delete)
productId	UUID	No	Foreign Key → Product.id
quantity	Int	No	Quantity ordered (≥ 1)
unitPrice	Decimal(10,2)	No	Frozen item price snapshot at creation time
notes	String	Yes	Custom item notes (e.g. "No spicy mayo")

8. Financial & Operational Models

Model: ExpenseType

Classifications for operational daily expenditures.

Column	Type	Nullable	Description
id	UUID	No	Primary Key
name	String	No	Category name (e.g. "Fresh Ingredients")
nameAr	String	No	Arabic name (e.g. "خامات وخضار طازج")
isActive	Boolean	No	Soft-activation status

Model: Expense

Individual operational petty cash disbursements.

Column	Type	Nullable	Description
id	UUID	No	Primary Key
expenseTypeId	UUID	No	Foreign Key → ExpenseType.id
userId	UUID	No	Foreign Key → User.id (Spender / recorder)
shiftId	UUID	Yes	Foreign Key → Shift.id
amount	Decimal(10,2)	No	Disbursed amount in EGP
description	String	No	Invoice or disbursement description
date	DateTime	No	Incurred date

Model: Shift

Operational work shifts.

Column	Type	Nullable	Description
id	UUID	No	Primary Key
cashierId	UUID	No	Foreign Key → User.id (Shift operator)
openedAt	DateTime	No	Shift start timestamp
closedAt	DateTime	Yes	Shift conclusion timestamp (null if active)
notes	String	Yes	Shift notes

Model: DailyClosing

Permanent shift reconciliation record.

Column	Type	Nullable	Description
id	UUID	No	Primary Key
shiftId	UUID	No	Foreign Key → Shift.id (Unique)
closedById	UUID	No	Foreign Key → User.id (Closing manager)
totalOrders	Int	No	Total non-cancelled orders count
cancelledOrders	Int	No	Total cancelled orders count
totalRevenue	Decimal(10,2)	No	Total gross revenue (Cash + Visa + Online)
totalCash	Decimal(10,2)	No	Total cash collected from orders
totalVisa	Decimal(10,2)	No	Total Visa collected at doorstep
totalOnline	Decimal(10,2)	No	Total pre-paid aggregator orders
totalDeliveryFees	Decimal(10,2)	No	Total net delivery fees collected
totalExpenses	Decimal(10,2)	No	Total cash operational expenses
netCash	Decimal(10,2)	No	Mandatory balance: totalCash - totalExpenses
actualCash	Decimal(10,2)	Yes	Physically counted cash in drawer
difference	Decimal(10,2)	Yes	Cash variance (actualCash - netCash)
notes	String	Yes	Closing discrepancy explanation
closedAt	DateTime	No	Closing timestamp

9. Security & Compliance Models (AuditLog)

Model: AuditLog

Immutable compliance log tracking every business mutation.

Column	Type	Nullable	Description
id	UUID	No	Primary Key
userId	UUID	No	Foreign Key → User.id (Actor)
action	AuditAction	No	Action type classification
entityType	String	No	Target entity (e.g. Order , User , Expense)
entityId	String	No	Primary key or order number of target entity
oldValue	Json	Yes	Snapshot of entity state prior to mutation
newValue	Json	Yes	Snapshot of entity state following mutation
timestamp	DateTime	No	Immutable timestamp (now())

10. Database Indexes & Performance Optimizations

To maintain sub-500ms analytical queries and instantaneous POS lookups under heavy concurrent traffic, the database schema implements the following B-Tree indexes:

```
-- Fast Customer Lookup by Phone
CREATE UNIQUE INDEX "Customer_phone_key" ON "Customer"("phone");

-- Fast Order Number Lookups & Sequence Resolution
CREATE UNIQUE INDEX "Order_orderNumber_key" ON "Order"("orderNumber");
CREATE INDEX "Order_status_createdAt_idx" ON "Order"("status", "createdAt");
CREATE INDEX "Order_brandId_platformId_idx" ON "Order"("brandId", "platformId");
CREATE INDEX "Order_customerId_idx" ON "Order"("customerId");
CREATE INDEX "Order_shiftId_idx" ON "Order"("shiftId");

-- Audit Log Filtering Performance
CREATE INDEX "AuditLog_timestamp_idx" ON "AuditLog"("timestamp" DESC);
CREATE INDEX "AuditLog_entityType_entityId_idx" ON "AuditLog"("entityType", "entityId");
CREATE INDEX "AuditLog_userId_idx" ON "AuditLog"("userId");

-- Catalog & Menu Sort Order Performance
CREATE INDEX "Category_brandId_sortOrder_idx" ON "Category"("brandId", "sortOrder");
CREATE INDEX "Product_categoryId_isActive_idx" ON "Product"("categoryId", "isActive");
```

Official Database Schema & Data Dictionary for Order Control System. Maintained by Database Engineering.